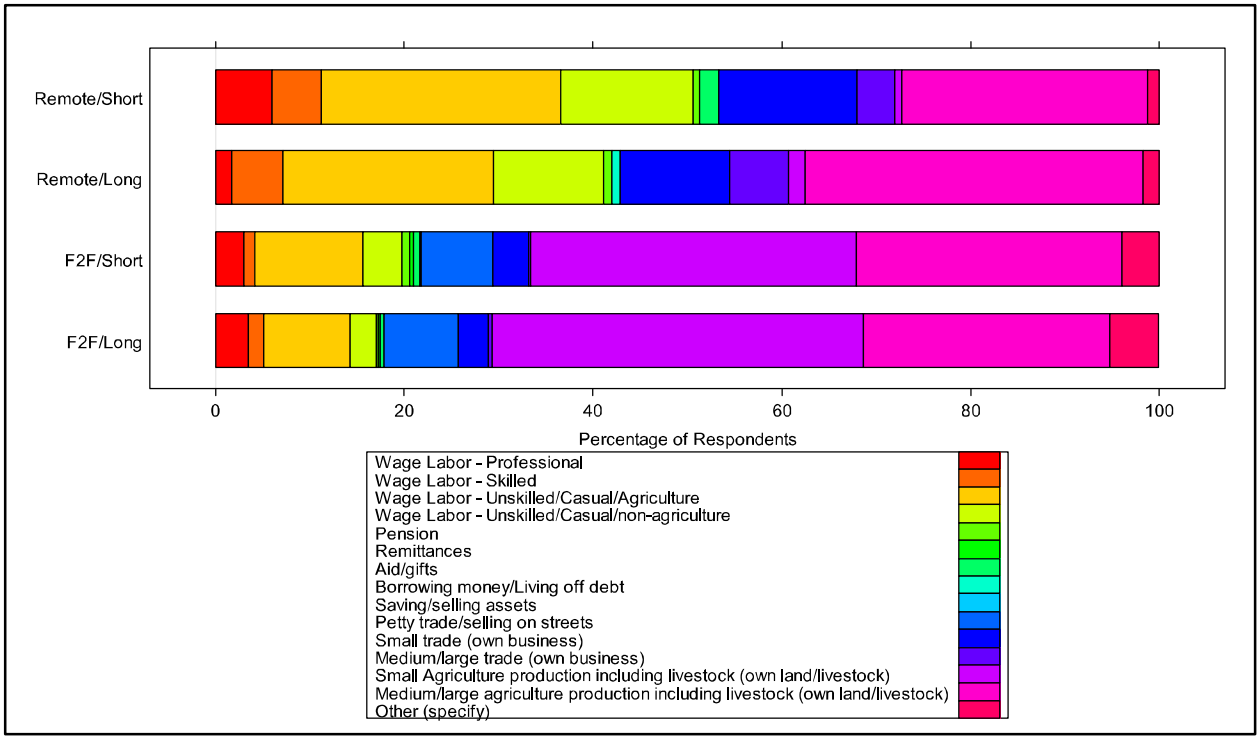
Comments on rCARI/CARI comparison report for the Central African Republic

Overall, the report is very well written and the experiment is described at an excellent level of detail. My comments concern the conclusions regarding differences related to modality, and I'm sharing some brief analysis in four specific areas.

Point 1: Although the report indicates few differences between the F2F and Remote respondents with respect to demographic factors, there are notable differences among the groups with respect to livelihood. In particular, while over a third of the F2F respondents worked in small agricultural production, this proportion declined to less than 2% for the remote participants. Conversely, the percentage of respondents engaged in unskilled wage labor (either agricultural or non-agricultural) increased from 12% (F2F/long form) and 16% (F2F/short form) to 34% (remote/long form) and 39% (remote/short form). These differences are not only statistically significant, but they have a direct impact on the food security distributions in each respondent group (see Point 4).



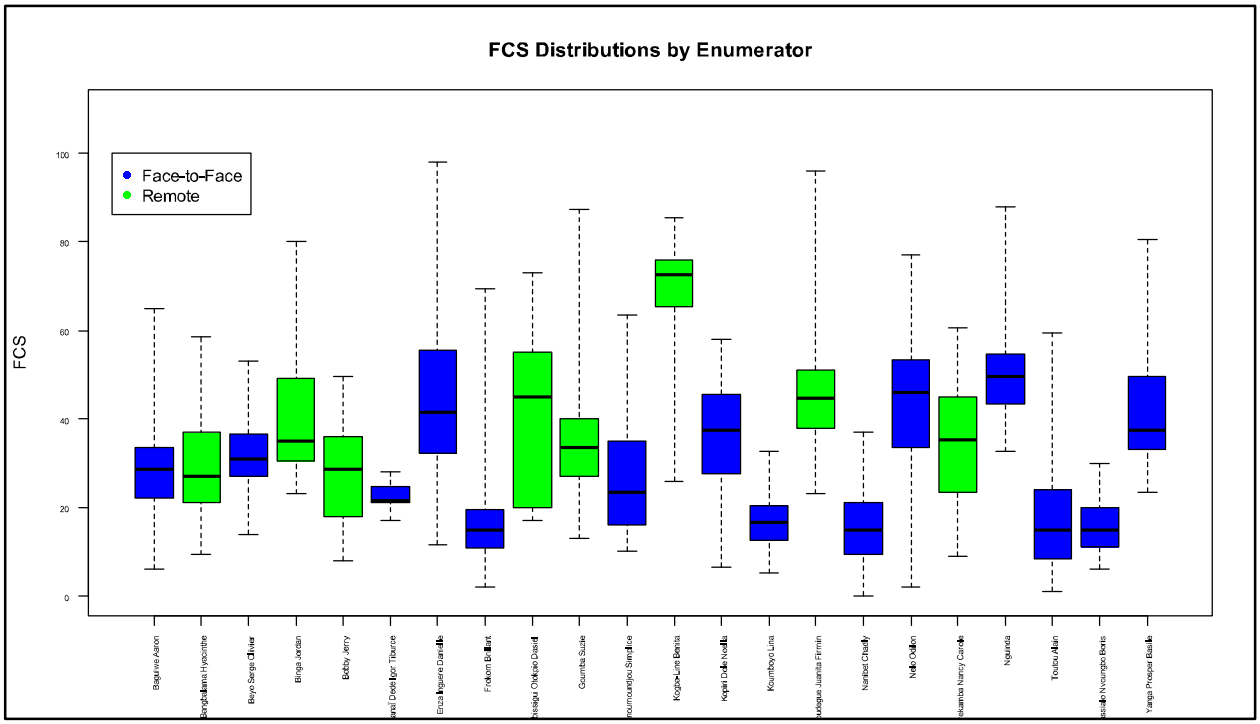
Point 2: The difference between the Income-based and FES indicator classifications is not consistent across all livelihoods. The most extensive differences were observed for small and medium agricultural producers, whose income is grouped in the "Acceptable" category (highlighted in "green" in the tables below). 80% of the 696 households listing agricultural production as their primary income source were assigned a less secure FS category using the Income-based indicator, and these accounted for 69% of all households that had an FES category that was two or more levels higher than their estimated category based on income. Notably, this result was consistent when either the long or short form of the FES module was employed.

F2F with long form FES					
Income Group	Inc<FES	Inc=FES	Inc>FES	Percentage of Sample	Percentage of Income Group with CARI_Inc < CARI_FES
Wage Labor - Professional	12	3	1	3.5%	75.0%
Wage Labor - Skilled	7	0	1	1.7%	87.5%
Wage Labor - Unskilled/Casual/Agriculture	28	4	10	9.1%	66.7%
Wage Labor - Unskilled/Casual/non-agriculture	8	2	3	2.8%	61.5%
Pension	1	0	0	0.2%	100.0%
Remittances	1	0	0	0.2%	100.0%
Aid/gifts	0	1	1	0.4%	0.0%
Petty trade/selling on streets	20	8	8	7.8%	55.6%
Small trade (own business)	13	2	0	3.2%	86.7%
Medium/large trade (own business)	2	0	0	0.4%	100.0%
Small Agriculture production including livestock	**158**	**19**	**5**	**39.4%**	**86.8%**
Medium/large agriculture production including livestock	**87**	**25**	**8**	**26.0%**	**72.5%**
Other (specify)	11	5	8	5.2%	45.8%
Total	348	69	45	100.0%	75.3%

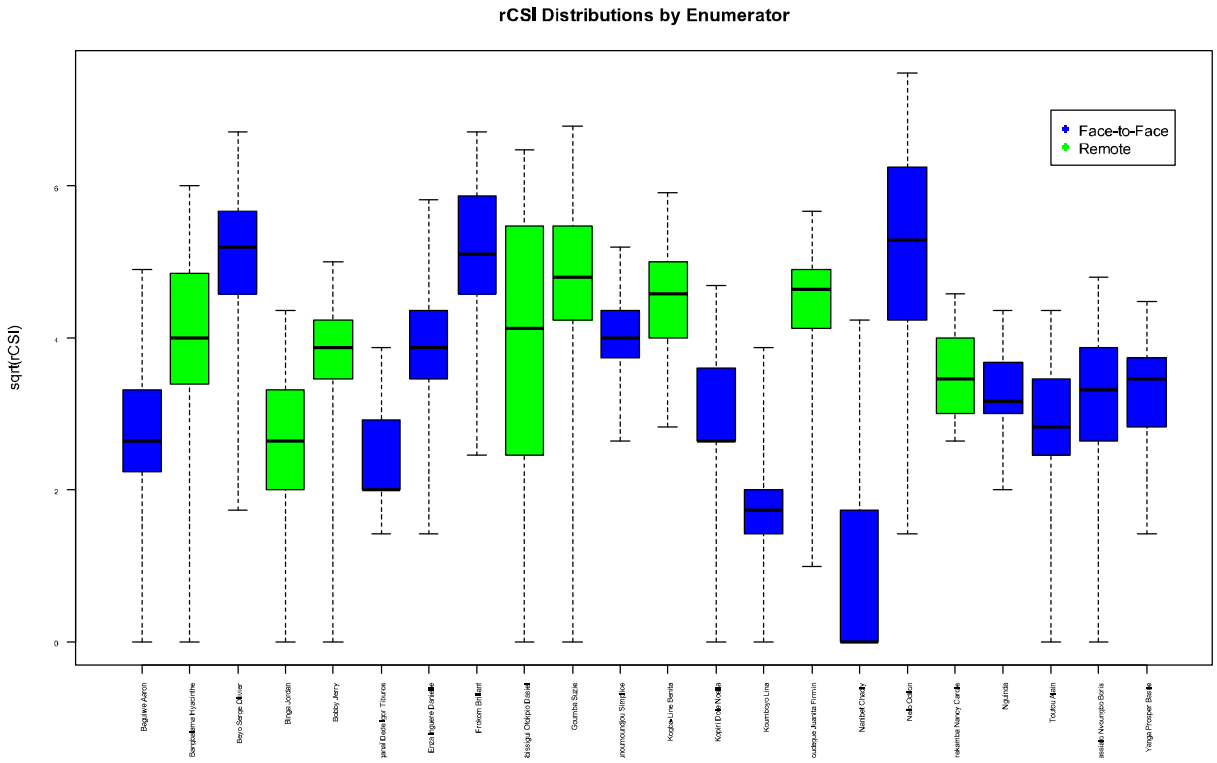
F2F with short form FES					
Income Group	Inc<FES	Inc=FES	Inc>FES	Percentage of Sample	Percentage of Income Group with CARI_Inc < CARI_FES
Wage Labor - Professional	10	5	0	3.0%	66.7%
Wage Labor - Skilled	5	1	0	1.2%	83.3%
Wage Labor - Unskilled/Casual/Agriculture	42	7	8	11.4%	73.7%
Wage Labor - Unskilled/Casual/non-agriculture	14	2	5	4.2%	66.7%
Pension	3	1	0	0.8%	75.0%

Remittances	2	0	0	0.4%	100.0%
Aid/gifts	0	1	2	0.6%	0.0%
Saving/selling assets	0	0	1	0.2%	0.0%
Petty trade/selling on streets	25	8	5	7.6%	65.8%
Small trade (own business)	14	5	0	3.8%	73.7%
Medium/large trade (own business)	1	0	0	0.2%	100.0%
Small Agriculture production including livestock	**141**	**28**	**3**	**34.5%**	**82.0%**
Medium/large agr production including livestock	**110**	**23**	**7**	**28.1%**	**78.6%**
Other (specify)	5	8	7	4.0%	25.0%
Total	372	89	38	100.0%	74.5%

Point 3: Nearly all the difference in FCS and rCSI scores between the F2F and Remote surveys can be explained by enumerator variation. There were 22 enumerators, 14 of whom conducted the F2F surveys while the remaining 8 conducted the remote surveys. As shown in the figure below, there was a high degree of variation associated with enumerators in both the Face-to-Face and remote surveys, with median FCS scores ranging from 15 to 72. ***A fitted ANOVA model predicting FCS as a function of enumerator alone explained 54% of the variability in scores. Furthermore, after accounting for enumerator variability using random effects, a model to predict FCS as a function of modality indicated that the modality effect was no longer statistically significant at the 0.05 level.***



Similarly, the variation in rCSI scores due to enumerator ID was far greater than any effects related to modality. As shown in the figure below, which plots sqrt(rCSI) by enumerator (to reduce skewness), enumerator variability far outweighs any potential bias due to modality effects. ***Enumerator effects explained 58% of the variation in sqrt(rCSI), and the modality effect was again insignificant after accounting for enumerator variation in a mixed effects model.***



Point 4: The difference in livelihood distributions between the F2F and remote surveys contributes to differences in LCS distributions by modality. As discussed in Point 1, the percentage of survey respondents involved in small agricultural production declined from 37% of the F2F sample to only 1% of the remote sample, while the percentage of respondents involved in unskilled wage labor increased from 14% in the F2F to 37% in the remote sample. As shown in the table below, wage laborers reported significantly higher levels of crisis and emergency coping (20.5% and 14.2%, respectively) than small agricultural producers (7.6% and 4.5%). While this does not rule out the possibility of modality-related effects, it is a confounding factor that needs to be considered since livelihood coping is, by definition, a function of a household's livelihood.

Income Group	No Livelihood Coping (Count)	Stress Coping (Count)	Crisis Coping (Count)	Emergency Coping (Count)	Total	% F2F Respondents	% Remote Respondents	Stress Coping (%)	Crisis Coping (%)	Emergency Coping (%)
Wage Labor - Professional	11	31	4	1	42	3.2%	4.2%	73.8%	9.5%	2.4%
Wage Labor - Skilled	7	21	5	4	28	1.5%	5.3%	75.0%	17.9%	14.3%

Wage Labor - Unskilled/Casual/Agriculture	59	95	34	23	162	10.3%	24.0%	58.6%	21.0%	14.2%
Wage Labor - Unskilled/Casual/non-agriculture	17	46	15	11	68	3.5%	13.0%	67.6%	22.1%	16.2%
Pension	0	7	1	1	7	0.5%	0.8%	100.0%	14.3%	14.3%
Remittances	1	2	1	0	3	0.3%	0.0%	66.7%	33.3%	0.0%
Aid/gifts	3	4	2	2	8	0.5%	1.1%	50.0%	25.0%	25.0%
Borrowing money/Living off debt	0	1	1	0	1	0.0%	0.4%	100.0%	100.0%	0.0%
Petty trade/selling on streets	15	58	9	3	74	7.7%	0.0%	78.4%	12.2%	4.1%
Small trade (own business)	14	52	13	6	69	3.5%	13.4%	75.4%	18.8%	8.7%
Medium/large trade (own business)	1	13	4	3	16	0.3%	5.0%	81.3%	25.0%	18.8%
Small Agriculture production	148	200	27	16	357	36.8%	1.1%	56.0%	7.6%	4.5%
Medium/large Agriculture production	65	248	110	33	339	27.1%	30.2%	73.2%	32.4%	9.7%
Other (specify)	9	39	12	3	48	4.6%	1.5%	81.3%	25.0%	6.3%